Chapter B6: Life on Earth – past, present and future

Overview

The modern explanation of evolution by natural selection is one of the central ideas in biology. The historical development of the explanation and its journey to widespread acceptance in the science community illustrate key *Ideas about Science*.

Learners explore ideas about evolution in Key Stages 2 and 3, so by GCSE (9–1) they should be familiar with the concepts of variation (at phenotype level), adaptation, advantage, competition and natural selection. In Topic B6.1, learners begin to expand their understanding by linking variation to genetics, and the concept of evolution by natural selection is explored within the story of how the theory was developed, evaluated and modified by the scientific community. The topic considers the importance of evidence as the basis for widespread scientific acceptance of the theory, and probes reasons why some people may still not accept it.

The effects that sexual and asexual reproduction have on evolution are considered in Topic B6.2, followed by a brief examination in Topic B6.3 of the impact that developments in scientific understanding have had on the way we classify the diversity of life on Earth today.

Finally, in Topic B6.4 learners examine the impacts of human activities on the Earth's biodiversity, the tremendous importance of protecting it, issues that affect decision making, and ways in which our understanding of science can help us to interact positively with ecosystems so that biodiversity and ecosystem resources are conserved for the future.

Learning about evolution and biodiversity before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- know that there are many different types of organisms living in many different environments, and that there are similarities and differences between all organisms
- recognise that living organisms can be grouped and classified in a variety of ways based on commonalities and differences
- be able to use classification keys
- recognise that living organisms have changed over time and that fossils provide information about organisms that lived millions of years ago
- appreciate that organisms live in habitats to which they are adapted
- recognise that organisms produce offspring of the same kind, but normally offspring vary and are not identical to their parents
- know that there is variation between individuals within a species, and that variation can be described as continuous or discontinuous
- understand that the variation means some organisms compete more successfully, resulting in natural selection
- appreciate that variation, adaptation, competition and natural selection result in the evolution of species
- understand that changes in the environment may leave organisms less well adapted to compete successfully and reproduce, which can lead to extinction
- be familiar with some of the reasons why it's important to protect and conserve biodiversity, and some ways of doing so.

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B6.1 How was the theory of evolution developed?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
The modern theory of evolution by natural selection combines ideas about genes, variation, advantage and competition to explain how the inherited characteristics of a population can change over a number of generations. It includes the ideas that: Mutations in DNA create genetic variants, which may be inherited. Most genetic variants do not affect phenotype, but those that do may increase an organism's ability to survive in its environments and compete for resources (i.e. confer an advantage). Individuals with an advantage are more likely to reproduce; thus, by natural selection, the proportion of individuals possessing beneficial genetic variants increases in subsequent generations. A new species can arise if the organisms in a population evolve to be so different from their ancestors that they could no longer mate with them to produce fertile offspring. Speciation is more likely to occur when two populations of an organism are isolated.	1. state that there is usually extensive genetic variation within a population of a species 2. recall that genetic variants arise from mutations, and that most have no effect on the phenotype, some influence phenotype and a very few determine phenotype 3. explain how evolution occurs through natural selection of variants that give rise to phenotypes better suited to their environment 4. explain the importance of competition in a community, with regard to natural selection 5. describe evolution as a change in the inherited characteristics of a population over a number of generations through a process of natural selection which may result in the formation of new species	
Charles Darwin noticed that the selective breeding of plants and animals had produced new varieties with many beneficial characteristics, quite different to their wild ancestors. Most of what we eat, and our ability to feed the growing human population depends on selectively bred plants and animals. Darwin wondered whether a similar process of selection in nature could have created new species.	6. explain the impact of the selective breeding of food plants and domesticated animals	

B6.1 How was the theory of evolution developed?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	***Linked learning opportunities***
The theory of evolution by natural selection was developed to explain observations made by Darwin, Wallace and other scientists, including: • the production of new varieties of plants and animals by selective breeding • fossils with similarities and differences to living species • the different characteristics shown by isolated populations of the same species living in different ecosystems. The development of the theory is an example of how scientists develop explanations. Darwin: made observations of the natural world; suggested natural selection to explain differences between fossils and living organisms, and between isolated populations; used ideas from Wallace and other scientists to improve his explanation; and shared his explanation with the scientific community (IaS3).	7. describe how fossils provide evidence for evolution 8. describe the work of Darwin and Wallace in the development of the theory of evolution by natural selection *(separate science only)*	**Ideas about Science:** • the theory of evolution by natural selection as an example of how scientific explanations are developed (IaS3)
The theory of evolution by natural selection illustrates how scientists continue to test a proposed explanation by making new observations and collecting new evidence, and how if the explanation is able to explain these it can become widely accepted by the scientific community (IaS3). For example, the spread of antibiotic resistance in bacteria can be explained by mutation, advantage and natural selection.	9. describe modern examples of evidence for evolution including antibiotic resistance in bacteria	**Ideas about Science:** • the theory of evolution by natural selection as a scientific explanation modified in light of new observations and ideas (IaS3)
Most scientists accept the modern theory of evolution because it is the best explanation for many of our observations of the natural world. However, some people do not accept it either because they are unaware of (or do not understand) the evidence, or because it does not fit with their beliefs (IaS4).	10. explain the impact of these ideas on modern biology and society *(separate science only)*	**Ideas about Science**: • reasons why different people do or do not accept an explanation (IaS4)

B6.2 How do sexual and asexual reproduction affect evolution? *(separate science only)*

Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
The evolution of a population or species is affected by whether the individual organisms reproduce sexually or asexually. Sexual reproduction occurs at a slower rate than asexual reproduction, but provides genetic variation in the offspring.	1. explain some of the advantages and disadvantages of asexual and sexual reproduction in a range of organisms	

B6.3 How does our understanding of biology help us classify the diversity of organisms on Earth?

Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
The enormous diversity of organisms on Earth can be classified into groups on the basis of observed similarities and differences in their physical characteristics and, more recently, their DNA. We are more likely to classify species into the same group if there are lots of similarities in their genomes (i.e. if they have many genes, and genetic variants, in common). Genome analysis can also suggest whether different groups have a common ancestor, and how recently speciation occurred.	1. describe the impact of developments in biology on classification systems, including the use of DNA analysis to classify organisms	

B6.4 How is biodiversity threatened and how can we protect it?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
The biodiversity of the Earth, or of a particular area, is the combination of the diversity of living organisms, the diversity of genes these organisms have, and the diversity of ecosystems. The biodiversity of many areas is being reduced by activities related to increasing human population size, industrialisation and globalisation. Such interactions can result in ecosystems being damaged or destroyed, populations dying out, and species becoming extinct when conditions change more quickly than they can adapt. Humans can interact with ecosystems positively by using ecosystem resources in a sustainable way (at the same rate as they can be replaced), and by protecting and conserving biodiversity. All organisms, including humans, depend on other organisms and the environment for their survival. Protecting and conserving biodiversity will help ensure we can continue to provide the human population with food, materials and medicines. Biodiversity can be protected at different levels, including protection of individual species, protection of ecosystems, and control of activities that contribute to global climate change. Decisions about protecting and conserving biodiversity are affected by ecological, economic, moral and political issues (IaS4).	1. describe both positive and negative human interactions within ecosystems and explain their impact on biodiversity **2. evaluate evidence for the impact of environmental changes on the distribution of organisms, with reference to water and atmospheric gases *(separate science only)*** 3. describe some of the biological factors affecting levels of food security including increasing human population, changing diets in wealthier populations, new pests and pathogens, environmental change, sustainability and cost of agricultural inputs *(separate science only)* 4. explain some of the benefits and challenges of maintaining local and global biodiversity 5. extract and interpret information related to biodiversity from charts, graphs and tables M2c, M4a 6. describe and explain some possible biotechnological and agricultural solutions, including genetic modification, to the demands of the growing human population *(separate science only)*	**Specification links:** • greenhouse gases and global warming (P1.3, C1.1) **Ideas about Science:** • the impacts of science on biodiversity, including negative impacts and potential solutions (IaS4) • decision making in the context of the protection and conservation of biodiversity (IaS4) **Practical work:** • measure living and non-living indicators to assess the effect of pollution on organisms